Muhammad Umer Rizwan

Also known as Umer Lakhani

Computer science student & developer · seeking software engineering & frontend internships

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PROFILE

I like to understand why a solution works — not only that it does. I enjoy breaking complex problems into manageable parts and keeping code readable. Faculty have recognised my sense of responsibility and teamwork, and I care about being dependable and asking good questions.

EDUCATION

BSCS, GU TECH, Karachi 2026 — 2030 (expected)

Undergraduate computer science studies.

Intermediate · Computer Science, Govt. Degree Boys College Gulistan-e-Johar Aug 2024 — Jun 2026

C++, object-oriented programming, and relational database systems.

Matriculation, Bait-Ul-Ilm Secondary School Aug 2019 — May 2024

Computer science foundations, programming logic, and procedural C.

SKILLS

Programming	C, C++, Python, Object-oriented programming
Core computer science	DBMS, Data structures, Algorithms, Memory basics
Workflow	Problem decomposition, Modular design, Debugging
Frontend	HTML, CSS, JavaScript (learning)
Design & research	Technical research, Requirements, Accessibility basics

PROJECTS

EduTrack Academic Management C++ · OOP

A console academic management system in C++17: student and faculty classes, file-based record persistence, input validation and credit-weighted GPA calculation, with 34 automated checks.

github.com/muhammadumera104-boop/portfolio/tree/main/case-studies/edutrack

AlgoVisualizer & Pathfinding Engine Python · Algorithms

A Python terminal tool that draws BFS, DFS, Dijkstra and A* on a grid with mud, animates the search, and generates repeatable mazes. Standard library only, with 15 tests.

github.com/muhammadumera104-boop/portfolio/tree/main/case-studies/algovisualizer

Custom Memory Allocator & DS Toolkit C · Systems

A malloc and free replacement in C on a fixed 64 KiB heap: first fit, block splitting, coalescing, misuse detection and a heap consistency checker, plus a growing array and a stack built on it. 11,234 automated checks.

github.com/muhammadumera104-boop/portfolio/tree/main/case-studies/allocator

GU TECH Learning Management System React · Node.js · Supabase

A role-based learning management system. Admins approve accounts and publish courses; teachers run lessons, assignments, grading and server-scored timed quizzes; students learn, submit work and chat. Row-level security on every table, private file storage and 59 automated API tests. Built with AI assistance.

github.com/muhammadumera104-boop/university-portal

MindBridge HTML

A private phone companion for students: decide before you open the phone, get a warning before time is up, wind down at night, and always know where a real person is.

github.com/muhammadumera104-boop/digital-well-being

Also built (small browser apps with live demos on the portfolio): MediQueue, StudySpace, Quizora, Noir & Bean, Wanderlust.

RECOGNITION AND LANGUAGES

Hafiz-e-Quran · Wifaq-Ul-Madaris. Completed memorisation of the Holy Quran, verified by Wifaq-Ul-Madaris.

Urdu · English · Native · Fluent. Comfortable communicating and collaborating in both.